

Trigger Mechanism Calculus

You perturb one trigger — an initial condition — and the policy fails. Eight cases stand between that observation and the sentence “mechanism m caused it.” Each case is decided below by a theorem, not by a plot.

Worked against GR00T N1.7 · LIBERO-Plus · runs/embed_derisk

Reads: Grant, Zhao & Wang, arXiv:2603.19233 2026-09-22

What is proved here, and what is not

The paper you sent is empirical. Its numbers — 97% → 0% on GR00T under a null prompt, 0% cross-task injection success, 28–92% zero-effect ablation rates — are measurements, and no proof makes a measurement true. What *is* provable is the calculus that turns such measurements into claims about mechanism: which inferences are licensed, which are blocked, and what an experiment must look like before its null result means anything.

That calculus is what you need. Your pipeline is: perturb a trigger → find the dependent mechanisms → locate the failing one → synthesise refined data → retrain → check improvement. Every arrow in that chain is an inference that can silently fail. Below, each is a theorem with a proof, and each is anchored to a result in the paper it licenses or restricts — and, where your repo has already run the experiment, to what you measured.

STANDING ASSUMPTION

The paper studies GR00T N1.5; you run N1.7. Theorems 1–10 are architecture-agnostic and hold for both. Every *instantiation* to N1.7 below is derived from your own reading of `action_head/cross_attention_dit.py:357–380`, not from the paper. Where the paper's N1.5 number is used as an anchor it is marked **measured**, never as a premise.

Definitions

D1 (Computation graph). A policy is a finite DAG $G = (M, E)$ whose nodes M are *mechanisms*. Each node computes $h_m = f_m(h_{pa(m)}; \theta_m)$ from its parents. Source nodes are input channels; one sink is the action A .

D2 (Trigger). A trigger T is a variable you control that enters the graph at a known set of source nodes. A *perturbation* is a pair (t, t') of its values. An initial condition enters at the camera and at proprioception; a prompt rewrite enters at the text channel; a light change enters at the camera alone.

D3 (Intervention). $A[do(h_S \leftarrow v)]$ is the action obtained by running the graph forward with the activations of $S \subseteq M$ forced to v . *Injection* (the paper's term) sets v to the activations recorded on another episode; *ablation* sets it to zero or to a dataset mean.

D4 (Dependence). m is *trigger-dependent* if $h_m(t) \neq h_m(t')$ for some perturbation. Write $D(T)$ for the set of such mechanisms.

D5 (Locus). m is a *locus of failure* for (t, t') if there exists v with $A[do(h_m \leftarrow v)]$ succeeding at t' while the unintervened run fails. Write $L(T)$.

D6 (Cut set). S is a cut set for T if every directed path from a T -source to A contains a node of S .

D4 and D5 are the two things routinely conflated in failure analysis, and Theorem 2 shows they are different sets. Almost every wasted experiment in this area measures $D(T)$ and reports it as $L(T)$.

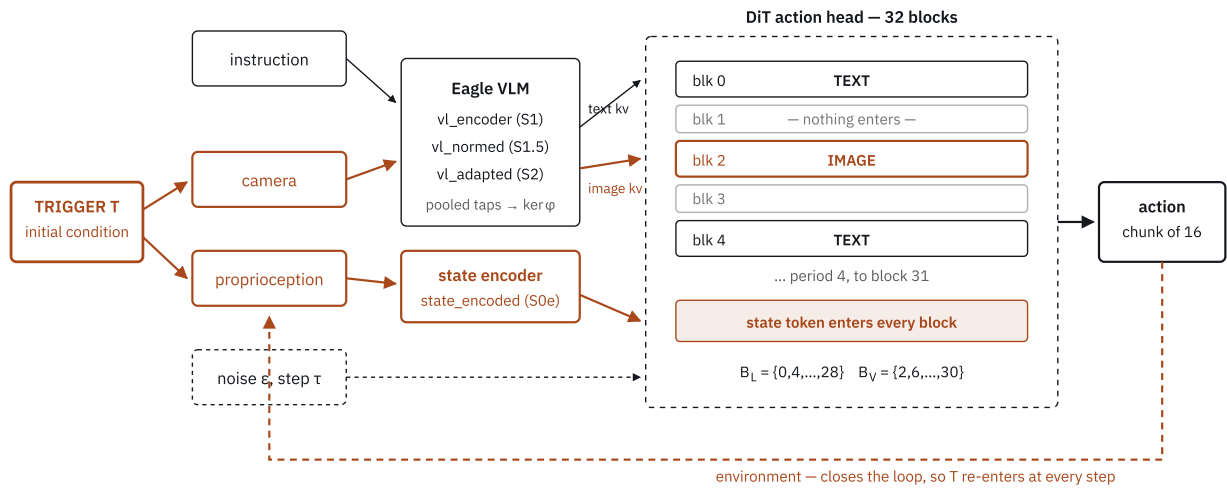


Figure 1. The graph an initial-condition perturbation actually moves through. Oxide marks every edge reachable from T ; grey marks what it cannot reach in one forward pass. Block indices are read from N1.7's forward pass: odd blocks receive no conditioning at all, text enters at $idx \bmod 4 = 0$, image at even blocks with $idx \bmod 4 = 2$. The dashed return edge is why Theorem 7 matters: T is not injected once, it is re-injected 500 times.

PART I — WHICH MECHANISMS DEPEND ON THE TRIGGER

Cases 1–5

THEOREM 1 · REACHABILITY — CASE 1

If no directed path runs from a T -source to m , then $h_m(t) = h_m(t')$ for every perturbation of T . Hence $D(T) \subseteq desc(T)$, and the inclusion is computable from the architecture alone, before a single rollout.

PROOF. Induct on a topological order of G . If m is a source and not a T -source, h_m is a constant of t . Otherwise every parent $p \in pa(m)$ is also outside $desc(T)$ — else a path through p would reach m — and by hypothesis each h_p is constant in t . Since f_m is a function of its parents and θ_m , h_m is constant in t . $\square$

Trivial to prove, and it is the most expensive theorem to ignore, because it prices an experiment before you run it. Instantiated on Figure 1, for a single forward pass:

PERTURBED TRIGGER	REACHES EAGLE VL	REACHES STATE ENC.	REACHES DIT TEXT BLOCKS <i>DIRECTLY</i>	CHEAPEST VALID PROBE
robot initial joint state	no [†]	yes	no	state_encoded
object initial pose	yes	no	no	per-token VL, blocks B_V
camera pose / lighting	yes	no	no	per-token VL, blocks B_V
instruction rewrite	yes	no	yes	blocks B_L only

[†] Not in the first forward pass. It reaches the VL encoder from step 2 onward through the closed loop of Figure 1, with the arm in a different place in frame. The distinction is the whole content of Corollary 1.1.

WHAT R-036 ACTUALLY MEASURED

R-036 found every VL signal flat ($0.37\times$ – $1.08\times$, none significant) and only state_encoded separating, at $12.95\times$, $p = 4\times 10^{-6}$. The 2026-09-22 correction then established that the sample could not have tested the hypothesis: of 40 episodes, **Camera Viewpoints has $n = 2$** — the category the capture was aimed at — while Robot Initial States and Objects Layout together carry 23 episodes and 12 of the 18 failures.

Theorem 1 prices exactly that composition, and splits it in two. Robot-initial-state is adjacent to the state encoder and one closed-loop hop from Eagle, so its VL flatness is predicted *a priori* at $h=1$ and carries no information about the VL pathway. Objects Layout, by the same table, *does* reach the VL encoder — its flatness is Theorem 6's territory, not Theorem 1's. The gate was neither met nor failed because the estimand and the sample were disjoint, which is the right call to have made. The category-balanced re-draw is the experiment; Theorem 1 says which categories belong in which arm, and Theorems 6 and 6b say what the readout must be before the arms can be compared.

COROLLARY 1.1 · THE LOOP DEFEATS STATIC PRUNING AT $H > 1$

In the unrolled graph over horizon H , with the environment edge of Figure 1 present, $\text{desc}(T)$ reaches every mechanism at every step $h \geq 2$. Therefore Theorem 1 prunes only the *first* forward pass, and any pruning claim must carry the step index it holds at.

PROOF. The environment edge makes o_{h+1} a descendant of A_h , which is a descendant of every conditioning channel. Compose. $\square$

THEOREM 2 · DEPENDENCE $\neq$ LOCUS — CASES 2 AND 3

There exist graphs with $m \in D(T) \setminus L(T)$ (activations move, mechanism is innocent) and graphs with $m \in L(T) \setminus D(T)$ (activations are perfectly flat, mechanism is the locus). Neither set contains the other.

PROOF. For $D \setminus L$: let $h_1 = T$ and $A = h_2$ with h_2 independent of h_1 . Then h_1 moves with every perturbation and no intervention on it changes A . For $L \setminus D$: let $h_1 = T \in \{0, 1\}$, let $h_2 \equiv c$ be a constant mechanism, and $A = h_2 \cdot (2h_1 - 1)$. Perturbing T leaves h_2 exactly flat, yet $\text{do}(h_2 \leftarrow -c)$ repairs the sign at t' . So $h_2 \in L(T)$ while $h_2 \notin D(T)$. $\square$

Case 3 is the one that bites. A flat probe is *not* an alibi: a mechanism whose output is constant can still be the thing that is wrong, because the failure lives in the *interaction* between its constant output and a trigger-shifted sibling. Concretely on Figure 1: an image-block key/value tensor that is identical across the perturbation can still be the locus if the state token it is mixed with has moved. Observational probing cannot see this; only an intervention that varies h_m can. This is the formal reason the paper's **measured** result that injecting GR00T activations transfers 57.0% of behaviour overall — 85.6% on *goal*, 33.3% on *long* — carries information that no amount of embedding geometry does.

THEOREM 3 · EXACT DECOMPOSITION OF THE TOTAL EFFECT

Fix a perturbation and an ordering π of $M = \{m_1, \dots, m_n\}$. Let $F(S) = A[\text{do}(h_S \leftarrow h_S(t))]$ evaluated on the t' run, and $S_i = \{m_{\pi(1)}, \dots, m_{\pi(i)}\}$. Then, with no assumptions beyond determinism,

$$TE = F(M) - F(\emptyset) = \sum_{i=1}^n [F(S_i) - F(S_{i-1})] \equiv \sum_i \Delta_i^\pi \quad (3.1)$$

PROOF. The sum telescopes. $\square$

PRECONDITION — DETERMINISM IS NOT FREE ON N1.7

Theorem 3 needs $F(S)$ to be a *function*: the term that enters one increment with a plus sign and the next with a minus sign must be the same number, or nothing cancels. That holds given (o, l, s, ε) — the noise draw is an input channel, which is why it appears as a source node in Figure 1. It does not hold given (o, l, s) alone, and N1.7's denoise loop starts from an **unseeded** `torch.randn` (`groot_n1_7.py:657`). As written, no patching sweep on this stack satisfies the premise.

Common random numbers. Fix ε once and reuse the same draw across source, target and every patched run. Nothing about the model changes; the noise is simply held constant alongside the seed and the initial state, and $F(S)$ is a function again. The alternative — defining $\bar{F}(S) = \mathbb{E}_\varepsilon[F(S)]$, which telescopes by linearity — is true but expensive: each increment becomes a difference of two Monte Carlo estimates, $\text{Var}(\Delta_i) \approx 2\sigma^2/k$, and the increments being read are small differences between adjacent prefixes. Shared ε is the variance reduction that makes the expectation route affordable, not an alternative to it. Theorem 7 sharpens the point: under the closed loop a different ε at step 1 diverges the whole trajectory, so two unseeded runs are not noisy copies of one episode — they are different episodes.

Both modes are needed, on separate paths. R-036 *relies* on the unseeded draw: it is what forces labels to come from the run being scored rather than being borrowed. Keep that for the OOD labelling protocol, and add a seeded mode for every intervention. Assert it rather than assume it — run $F(\emptyset)$ twice and require bitwise equality, which also catches the GPU-side nondeterminism a seed alone does not fix (atomics, kernel selection, bf16 reduction order).

THEOREM 3B · THE DECOMPOSITION IS ORDER-DEPENDENT — CASE 4

There is a graph and a perturbation for which Δ_i^π assigns the entire total effect to m_1 under one ordering and the entire total effect to m_2 under another. Therefore a layer-wise patching curve is not an attribution until its ordering is fixed and declared.

PROOF. Let $A = m_1 \wedge m_2$, target $(0,0) \rightarrow 0$, source $(1,1) \rightarrow 1$, so $TE = 1$. Under $\pi = (1,2)$: $F(\{m_1\}) = 1 \wedge 0 = 0$, so $\Delta_1 = 0$ and $\Delta_2 = 1$. Under $\pi = (2,1)$ the labels swap. $\square$

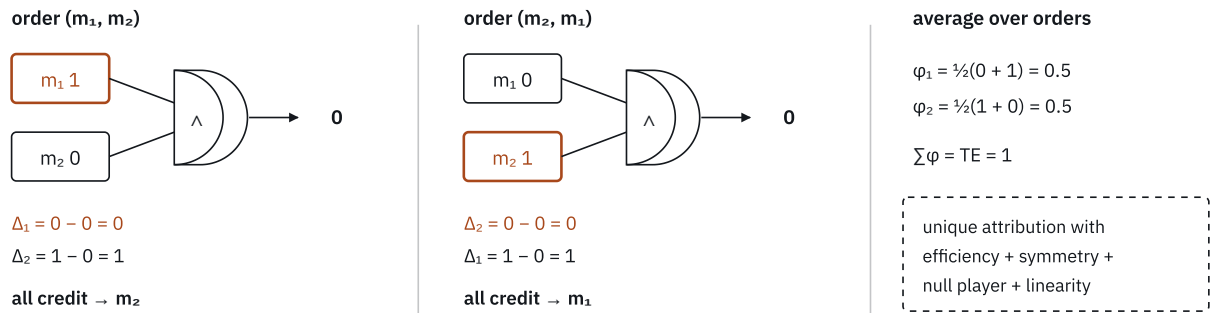


Figure 2. Theorem 3b in two gates. The same circuit, the same total effect of 1, and two opposite attributions. Averaging over all $n!$ orders — the Shapley value — is the unique fix satisfying the four standard axioms, and it costs 2^n forward passes, which is why the practical move is to fix one order and pre-register it.

THEOREM 4 · CUT-SET SUFFICIENCY — CASE 5

Let source and target episodes differ *only* in T . (i) If S is a cut set for T , then $A[do(h_S \leftarrow h_S(t))]$ = $A(t)$ exactly: injecting at S transfers the whole trigger effect. (ii) Conversely, if injection at S reproduces $A(t)$ for all such pairs, then $A \perp T \mid h_S - h_S$ is a sufficient statistic for the trigger, and everything downstream that is not in S carries no additional trigger information.

PROOF. (i) Induct downstream of S in topological order. Take $m \notin S$ with all parents already shown to equal their t -values. Each parent is either in S (forced to its t -value), or outside $desc(T)$ (equal by Theorem 1, since the episodes differ only in T), or already handled. So h_m equals its t -value; apply at A . Paths not through S do not exist by D6. (ii) Conditional on h_S , the map to A is by hypothesis invariant to which t produced the rest of the state; invariance of the conditional law is exactly conditional independence. $\square$

COROLLARY 4.1 · N1.7'S LANGUAGE CUT SET IS AN INDEX SET

In N1.7's DiT, text enters only at blocks $B_L = \{0, 4, 8, \dots, 28\}$ and image only at $B_V = \{2, 6, \dots, 30\}$; odd blocks receive no conditioning. Hence B_L is a cut set for the instruction channel and B_V for the visual channel, *within the head*. Two consequences: ablating any odd block cannot change the policy's language dependence, and the pathway separation the paper establishes for N1.5 by intervention is available in N1.7 **by index**, with no patching, no matched pairs, no seed control and no token alignment.

PROOF. Immediate from D6 applied to the block-conditioning schedule; the only route from the text key/value tensor into the residual stream is a cross-attention at a block of B_L . $\square$

COROLLARY 4.1 IS UNVERIFIED — AND ITS FALSIFIER

Status: derived from a reading of the forward pass, never confirmed by a measurement. The schedule was read from

action_head/cross_attention_dit.py:357–380, and an earlier reading of the same code gave period 2 — inferred from the config name `attend_text_every_n_blocks: 2` rather than from the loop, independently and wrongly, by two people. The current reading is period 4 because the counter runs over even blocks only. Until a layer-wise readout confirms it, every cut-set claim resting on it is provisional.

The structural null is not “no periodicity” — it is *periodicity at exactly the architectural phase*. If a layer-wise readout shows language structure peaking at phase $\neq 0 \bmod 4$, either the readout is phase-shifted or the cut set is wrong; both are bugs, not findings. A phase error here would have propagated silently, which is precisely what happened twice in your own repo before the forward pass was read.

THEOREM 5 · REDUNDANCY — CASE 6: A NULL ABLATION PROVES NOTHING

For every $n \geq 2$ and $1 \leq k \leq n-1$ there is a graph in which every one of n mechanisms is fully trigger-dependent, the trigger fully determines the action, and yet *every single-mechanism ablation has exactly zero effect*. More generally no subset of size $< n-k+1$ has any effect.

PROOF. Let $h_j = T \in \{0, 1\}$ for $j = 1 \dots n$ and $A = 1[\sum_j h_j \geq k]$. Every $h_j \in D(T)$, and $A = T$. Ablate any S with $|S| \leq n-k$: at $T = 1$ the surviving count is $n - |S| \geq k$, so $A = 1$ still; at $T = 0$ the count is 0 either way. Effect zero. $\square$

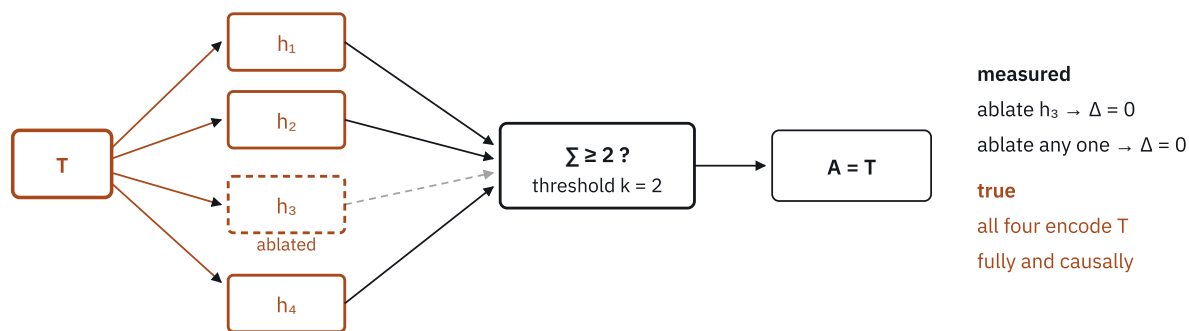


Figure 3. Theorem 5 drawn at $n=4$, $k=2$. Every single-unit ablation reads exactly zero while every unit carries the trigger. The paper's **measured** 28–92% zero-effect rates across 15,096+ concept–task pairs — 92% on the 4096-dim OpenVLA-OFT, 28% on the 480-dim SmolVLA — are therefore consistent with full encoding, and the width ordering is what redundancy predicts. Single-unit search has a provable blind spot; minimal sufficient sets must be searched over sets.

PART II — WHAT THE MEASUREMENT CAN SEE

Case 7, and two bounds on the evidence

Part I is about the network. Part II is about your instrument. Three results here, and the first of them settles the caveat you flagged yourself in R-036 — in your favour, and more sharply than the caveat was stated.

THEOREM 6 · CASE 7: PERMUTATION-INVARIANT READOUTS HAVE EXACTLY ZERO POWER AGAINST *WHERE*

Let a role's activation be a token matrix $H \in \mathbb{R}^{n \times d}$. Suppose the perturbation acts on tokens by a permutation, $H(t') = PH(t)$ for some permutation matrix P . Then for *every* permutation-invariant readout φ — mean-pool, max-pool, sum, any order statistic, any symmetric function, and any statistic ψ computed downstream of one —

$$\psi(\varphi(H(t'))) = \psi(\varphi(H(t))) \text{ pointwise, hence } \textit{power} = \textit{size} \text{ for any test at any } n. \quad (6.1)$$

PROOF. $\varphi(PH) = \varphi(H)$ by invariance, so the two random variables are equal as functions on the same sample space, not merely equal in distribution. Any test statistic is a measurable function of them and therefore identical; its rejection probability is the same under both hypotheses, which is the definition of power equal to size. Sample size does not enter. $\square$

Move an object across the table, or move the camera, and to first order the patch-token content is unchanged — it is *relocated*. That is the permutation case. `taps.py` stores `vl_encoder`, `vl_normed` and `vl_adapted` mean- and max-pooled across the token axis, and both poolings are permutation-invariant. So for the spatial component of the trigger a flat pooled VL signal is close to the only available outcome, and R-036's caveat 3 is not a 50/50 ambiguity to be resolved by more episodes. The measurement was well-run and correctly reported; the estimand sat largely in the kernel of the readout.

HOW EXACT IS THIS?

Theorem 6 is *exactly* true for an exact permutation, $H(t') = PH(t)$. Whether a real camera move or object displacement is a permutation of patch tokens is a modelling assumption, and an idealisation: real perturbations also change content — occlusion, shading on the moved object, new background patches. Under an *approximate* permutation the conclusion weakens from zero power to **attenuation in proportion to how close the perturbation is to a pure relocation**. A pooled null is therefore weak evidence, not no evidence — and the measured fact that pooled S2 separates at all in the first two forwards shows the non-relocation part of the change is not negligible here.

COROLLARY 6.1 · THE MINIMAL REPAIR — EQUIVARIANT READOUTS

Let $w_i = \|h_i\|_2$ over patch tokens at grid positions (r_i, c_i) , and define the first spatial moment

$$\mu = (\sum_i w_i (r_i, c_i)) / (\sum_i w_i). \quad (6.2)$$

Under a pure translation of content by δ on the grid, $\mu \rightarrow \mu + \delta$. So μ is equivariant, separates exactly the family Theorem 6 annihilates, and costs three extra floats per role per forward pass.

PROOF. Substituting $(r_i, c_i) \rightarrow (r_i, c_i) + \delta$ with weights unchanged adds $\delta \sum w_i / \sum w_i = \delta$. $\square$

THEOREM 6B · MODALITY MIXING ATTENUATES, AND CAN ONLY ATTENUATE

Suppose the stored readout pools over a sequence carrying both modalities, so that $\varphi = \alpha u + (1-\alpha)w$, where u is the image component with class means μ_0, μ_1 and within-class variance σ^2 , w is a class-independent text component with mean m , and the mixing ratio α varies across episodes independently of the class. Write $\Delta = \mu_1 - \mu_0$. Then the achievable effect size on φ is

$$d' = \mathbb{E}[\alpha] \Delta / \sqrt{(\mathbb{E}[\alpha]^2 \sigma^2 + \text{Var}(\alpha) (\bar{\mu} - m)^2)} \leq d = \Delta / \sigma, \quad (6.3)$$

with equality iff $\text{Var}(\alpha) = 0$, and d' strictly decreasing in $\text{Var}(\alpha)$. The bias is **toward the null**: mixing can dilute a real separation but cannot manufacture one. Hence a *positive* result measured on a mixed readout is conservative and may be believed; a *null* is uninterpretable.

PROOF. By independence, $\mathbb{E}[\varphi | c] = \mathbb{E}[\alpha] \mu_c + (1 - \mathbb{E}[\alpha])m$, so the between-class gap is $\mathbb{E}[\alpha] \Delta$ — shrunk by $\mathbb{E}[\alpha] < 1$. By the law of total variance, conditioning on α gives $\text{Var}(\varphi | c) = \mathbb{E}[\alpha]^2 \sigma^2 + \text{Var}(\alpha) (\mu_c - m)^2$, a sum of the original noise and a strictly non-negative term contributed by the mixing. Divide. The monotonicity and the equality condition are read off (6.3). $\square$

This is the second, independent defect in the same tap, and it is not Theorem 6.

`taps.py` pools with `t.mean(axis=1)` over the whole VL sequence, while `groot_n1_7.py:451` builds an `image_mask` that `AlternateVLDiT` uses to drive the two streams through *separate* attention masks — the model treats them as separate channels and the capture averages them into one vector. Instruction length ran 14 to 23 words across R-036's 40 episodes, so $\text{Var}(\alpha)$ is not small and it varies for reasons uncorrelated with what the camera saw. **The fix is already written and has never been run:** `CaptureSink.set_token_mask` and the `_img_ / _txt_` pooling branch landed after the de-risk capture, and the 40 episodes in `runs/embed_derisk/capture` carry neither those arrays nor `n_image_tokens`. So this defect costs one re-capture, not a code change. Theorem 6b also settles the direction of the error, which matters for how the existing numbers should be read — the flat VL rows are weaker evidence than they look, and the 12.95× on the state pathway is not inflated by this mechanism.

THEOREM 6C · TEMPORAL AVERAGING — THE SAME ALGEBRA, ON THE AXIS NOBODY CHECKS

Let an episode-level statistic average a per-forward score over F forwards, $S = (1/F) \sum_j s_j$, and suppose the discriminative content is confined to the first f_0 forwards: $\mathbb{E}[s_j|c] = \mu_c$ for $j < f_0$ and $= m$ otherwise, with $\text{Var}(s_j) = \sigma^2$ throughout. Then, against the same statistic computed on the first f_0 forwards alone,

$$d'/d = \sqrt{f_0/F}. \quad (6.4)$$

The forwards beyond f_0 contribute no signal and their own variance, and the attenuation is *unbounded* in F : a long enough episode hides any finite early effect. If moreover F differs systematically by class — failures running to a step cap while successes terminate on success — the two classes are attenuated by different factors and S is not merely diluted but **biased**.

PROOF. The class gap is $\mathbb{E}[S|1] - \mathbb{E}[S|0] = (f_0/F) \Delta$ with $\Delta = \mu_1 - \mu_0$, and $\text{Var}(S) = \sigma^2/F$, so $d' = (f_0/F) \Delta \cdot \sqrt{F}/\sigma = f_0 \Delta / (\sigma \sqrt{F})$. Restricting to the first f_0 gives gap Δ and variance σ^2/f_0 , so $d = \Delta \sqrt{f_0}/\sigma$. Divide. The bias claim follows because f_0/F is then class-dependent, so the two conditional means are shrunk toward m by different amounts. $\square$

COROLLARY 6.2 · REPORT THE PREFIX CURVE, NOT THE ENDPOINT

f_0 is not known in advance, so no single F is the right one. The estimand is the statistic *as a function of* F . Restricting to $F \leq \min_c F_c$ also makes the comparison length-matched, which removes the bias term above at no cost.

MEASURED — AND THIS WAS THE DEFECT THAT ACTUALLY BIT

Prefix-restricted re-analysis of R-036's 40 episodes (experiments/ood_prefix.py, pre-registered 2026-09-22-prefix-restricted-reanalysis.md), no new GPU time:

K (FIRST K FORWARDS)	STATE_ENCODED	VL_ADAPTED_MEAN (S2)	VL_NORMED_MEAN (S1.5)
2	4.48× (p=.053)	5.30× (p=.0089)	0.81× ns
3	5.81× (p=.0095)	2.14× ns	0.92× ns
5	7.94× (p=.00044)	0.61× ns	0.61× ns
18 (R-036)	12.95× (p=4e-6)	1.08× ns	flat

Every failure has *exactly* 18 forwards — all hit the step cap — against a success mean of 7.77 (Mann-Whitney $p = 1.6 \times 10^{-8}$), so the bias clause of Theorem 6c is live, not hypothetical. For $k \leq 5$ every episode contributes exactly k points and the comparison is length-matched. R-036's “every VL signal is flat” is a **property of averaging over 18 forwards, not of the representation**. Two opposite profiles: `state_encoded` accumulates (Corollary 7.1), `vl_adapted_mean` decays (Theorem 6c) — and only S2 does it, with S1 and S1.5 flat at every k . Status: **exploratory**. This is one result among ~108 correlated tests, found by looking after the fact; Bonferroni on $p=.0089$ gives 0.96. It is a hypothesis for the category-balanced capture, pre-registered at $k=2$, not a finding.

THE PAPER MEASURED THE SAME THEOREM FROM THE OTHER SIDE

Training SAEs on mean-pooled action-token activations reaches $R^2 > 0.95$ reconstruction and **0.4%** rollout success on $\pi_{0.5}$, against **70%** per-token. On GR00T's VL-SA pathway pooling *improves* explained variance from 83% to 99% while being the pathway whose ablations matter least. Reconstruction quality and causal power are not the same quantity, and pooling trades the second for the first. Their operational rule — process VLA activations per token, never mean-pooled — is Theorem 6 as a lab protocol, and it is the single cheapest change to your capture path.

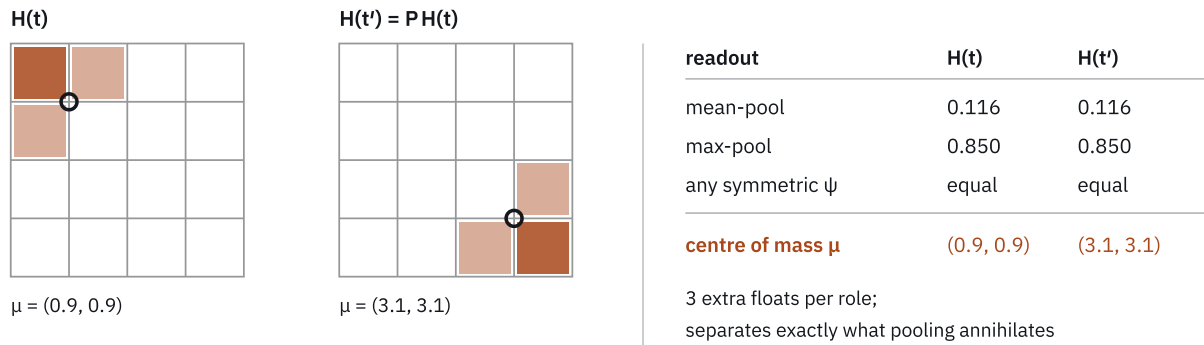


Figure 4. Theorem 6 in one picture. The circle marks the weighted centre of mass. Identical mean, identical max, identical every symmetric function — and the content is in a different place. Against an exact relocation a pooled tap has no power at all; against a realistic, approximate one it retains only what the non-relocation part of the change carries.

THEOREM 7 · HORIZON AMPLIFICATION (ROSS & BAGNELL, RESTATED)

Let π disagree with the demonstrator with probability ε per step under the demonstrator's own state distribution, with per-step cost in $[0,1]$ and horizon T . Then

$$J(\pi) \leq J(\pi^*) + T^2\varepsilon, \tag{7.1}$$

and the bound is tight in the worst case. Consequently a trigger that shifts the per-step distribution by a small ε can drive episode success to zero, and *episode-level* success differences are a quadratically amplified, saturating readout of a step-level effect.

PROOF. (Sketch, following the original.) Condition on the first step at which π deviates. Deviation first occurs at step i with probability at most ε ; after a deviation the state distribution is unconstrained and the remaining $T - i$ steps cost at most $T - i \leq T$ more than the demonstrator. Summing over $i = 1 \dots T$ gives $T \cdot \varepsilon \cdot T$. $\square$

COROLLARY 7.1 · POST-DIVERGENCE PROBES ARE NON-IDENTIFYING

Let h^* be the first step at which the policy diverges from the demonstrator. In the unrolled graph of Figure 1, every activation at any step $h > h^*$ is a descendant of A_{h^*} . Therefore an out-of-distribution score pooled over a whole episode separates failures from successes *whenever failure displaces the state at all*, with no trigger-specific mechanism required — and such a score has exactly zero power to distinguish “mechanism m carries the trigger” from “ m sits downstream of a failure that already happened.”

The remedy is a *prefix-restricted* score: use only forwards at steps $h \leq h^*$, with h^* estimated as the first divergence against the matched control that Theorem 8's pairing already gives you.

PROOF. The environment edge makes o_{h+1} a descendant of A_h ; compose along $h > h^*$. Failure means A_{h^*} left the demonstrator's state manifold, so all subsequent activations are evaluated off the reference distribution the OOD score was built from, and the score is by construction sensitive to that displacement. Since this holds for every failing episode regardless of which trigger caused it, the induced separation is constant across trigger families and carries no trigger information. $\square$

R-036'S PER-CATEGORY TABLE IS THIS COROLLARY

`state_encoded`'s failure rate is **broad across every category** — 27.8% camera, 34.7% layout, 52.8% noise and light, 60.4% robot-init — which disconfirmed the hypothesis, offered before the numbers were computed, that the 12.95 $\times$ would prove to be carried by the robot-init and layout episodes. Corollary 7.1 predicts exactly that breadth: a failing episode ends up displaced almost by definition, whatever moved it. The standing causation caveat in `RESULTS.md` is this theorem, and the prefix restriction is what converts the 12.95 $\times$ into a number that could mean something mechanistic — at no GPU cost, since the forwards are already captured and timestamped.

Two operational consequences. First, attribution must be done at the step level: the failing mechanism is identified by where the per-step action diverges, not by the binary at the end. Second, the suite ordering everyone observes — long-horizon suites worst — is partly a horizon artefact, not extra evidence about mechanism. The paper's GR00T cross-task displacement of 85.6% on *goal* against 33.3% on *long* (`measured`, 270 pairs) is consistent with an identical per-step effect under two different values of T , and your own `libero_10` sitting 34 points under

published is the same shape. Neither licenses a claim that long-horizon tasks use a different mechanism until the step-level effect is matched.

THEOREM 8 · PAIRING, AND WHAT IT BUYS

For a continuous per-episode readout with between-episode variance σ^2 and correlation ρ between the two arms of a matched pair, the paired-difference estimator has variance $2\sigma^2(1-\rho)/n$ against $2\sigma^2/n$ unpaired — a sample-size saving of $1/(1-\rho)$. For binary success with McNemar's test, detecting a discordance ratio $\psi = p_{01}/p_{10}$ at level α and power $1-\beta$ requires

$$m \geq (z_{\alpha/2} + z_{\beta})^2 (\psi+1)^2 / (\psi-1)^2 \text{ discordant pairs, } n = m / (p_{01}+p_{10}). \quad (8.1)$$

At $\psi=2$, $\alpha=0.05$, $\beta=0.2$: $m \geq 71$, and at a 25% discordance rate, $n \approx 284$ paired episodes. A trigger sweep that holds seed and initial state fixed across arms is therefore not a nicety; it is the difference between 284 episodes and a study that cannot resolve its own effect.

PROOF. Var of a difference of correlated variables is $\sigma^2+\sigma^2-2\rho\sigma^2$; divide by n . For (8.1), condition on the discordant count: under H_0 the split is Binomial(m , $\frac{1}{2}$) and under H_1 Binomial(m , $\psi/(\psi+1)$); the normal approximation to the two-proportion comparison gives the stated m . $\square$

YOU ALREADY HIT THIS

F9 reports the 256 vs 360 render comparison as unpaired (overall +5.7 pp, $p = 0.092$) while noting both runs used `seed=1000` with `init_states=true`, so the episodes *are* paired and McNemar would be tighter. Theorem 8 prices that: at a plausible ρ the paired test is the one that can resolve the effect, and the blocker is only that per-episode outcomes are not in `eval_info.json`. That is a logging change, not an experiment.

PART III — FROM MECHANISM TO DATA

Case 8, and what refined data has to be

Say Part I and Part II have done their work and you hold a mechanism. The plan is to synthesise refined data the pretraining never saw, retrain, and check for

improvement. Two theorems decide whether that plan can work at all — and they decide it before the GPU is booked.

THEOREM 9 • CASE 8: THE SHORTCUT THEOREM

Let the training distribution be $p(O, X, A)$ where X is an input channel (instruction, or any perturbation factor). Define the *channel-attributable risk gap* under squared loss

$$\Delta_X = \mathbb{E} \|\mathbb{E}[A \mid O, X] - \mathbb{E}[A \mid O]\|^2. \quad (9.1)$$

(i) If $X = g(O)$ almost surely, then $\Delta_X = 0$ and *some* risk-minimising policy ignores X entirely. (ii) More generally $I(A; X \mid O) = 0 \Rightarrow \Delta_X = 0$, and contrapositively $\Delta_X > 0 \Rightarrow I(A; X \mid O) > 0$. Hence a policy's insensitivity to a channel is a property of the *data*, not a defect of the architecture, and no amount of retraining on data with $\Delta_X = 0$ will create sensitivity to X .

PROOF. (i) If X is $\sigma(O)$ -measurable then $\sigma(O, X) = \sigma(O)$, so the two conditional expectations coincide a.s. and (9.1) vanishes; the policy $o \mapsto \mathbb{E}[A \mid O=o]$ attains the minimum. (ii) $I(A; X \mid O)=0$ is $A \perp X \mid O$, which gives $\mathbb{E}[A \mid O, X] = \mathbb{E}[A \mid O]$ a.s.; substitute. The converse of (ii) fails in general because (9.1) sees only conditional means, which is why Δ_X is the right estimand and mutual information is the certificate. $\square$

This is the whole of the paper's headline result, derived rather than observed. “The visual pathway dominates” is what Theorem 9(i) predicts for any benchmark whose scene determines the task: the instruction is a measurable function of the image, $\Delta_L = 0$, and behaviour cloning has no gradient pressure to read it. The paper's own suite table is the theorem's fingerprint — the same model swings from language-blind to language-critical with the *suite*, not the architecture:

MODEL / SUITE	BASELINE	NULL PROMPT	WRONG OBJECT	Δ_L IMPLIED
GROOT / goal	97%	0%	0%	large — scene fixes the objects, only text fixes the goal
GROOT / object	93%	70%	50%	partial
OpenVLA-OFT / goal	100%	10%	10%	large

MODEL / SUITE	BASELINE	NULL PROMPT	WRONG OBJECT	Δ_L IMPLIED
X-VLA / goal	94%	10%	4–10%	large
$\pi_{0.5}$ / object	77.4%	77.0%	74.2%	≈ 0 — vision alone identifies the task

All rows **measured**, arXiv:2603.19233 §4.4. $\pi_{0.5}$ across all prompt variations: $F(4, 3391) = 1.23$, $p = 0.247$, $\eta^2 = 0.012$.

The design rule this hands you. Refined data intended to make the policy sensitive to a trigger X must satisfy $I(A; X \mid O) > 0$ *within the new data*: the trigger must change the correct action while remaining un-inferable from everything else in the observation. Data that varies X in lockstep with a visual cue teaches nothing about X — the policy will read the cue. This is a checkable property of a dataset before training: estimate Δ_X by fitting two regressors, with and without the channel, on held-out episodes.

THEOREM 10 · NO EXCITATION, NO REPAIR — LINEAR MECHANISM

Let mechanism m compute $h = Wu + b$ from an input u , and let the refined dataset D have all its inputs in a subspace $U = \text{span}\{u_1, \dots, u_k\}$ with $U \neq \mathbb{R}^d$. Then for any differentiable loss, every gradient satisfies

$$\nabla_W \hat{R} \in \{v u^T : u \in U\}, \text{ so } W_t u = W_0 u \text{ for all } u \perp U, \text{ at every step } t. \quad (10.1)$$

The mechanism's response along any direction the data never excites is *exactly* its pretrained response, forever, under gradient descent from any initialisation.

Equivalently: the Fisher / Gauss-Newton matrix restricted to those directions is singular, the parameters are unidentified, and the minimum-norm solution leaves them at W_0 .

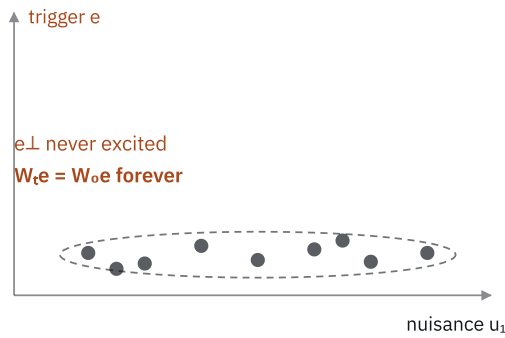
PROOF. For one sample, $\partial \ell / \partial W = (\partial \ell / \partial h) u^T$ by the chain rule, which is rank one with row space $\text{span}\{u\}$. The empirical gradient is a sum of such terms with $u \in U$, so its row space lies in U . Iterates stay in $W_0 + \{\text{row space} \subseteq U\}$, and for $u \perp U$, $(W_t - W_0)u = 0$. $\square$

Scope. This is proved for a mechanism that is *linear in its input*. The general nonlinear statement — gradients lie in the span of the realised Jacobian rows, so unexcited directions stay unidentified — is true in the same spirit but weaker, and in a deep residual network “never excited” is much harder to establish than it is here: skip connections and nonlinearities mix directions across layers, so a direction unexcited at one site may be excited at the next. Treat the linear case as the theorem and the rest as motivation, and establish non-excitation empirically per site rather than assuming it.

COROLLARY 10.1 · MATCHED COUNTERFACTUAL PAIRS ARE THE EFFICIENT DESIGN

Let e be the trigger direction you intend to repair. The learning rate along e is governed by $e^T \Sigma e$, where Σ is the input second-moment matrix of the refined data. Episode pairs identical in every respect except the trigger maximise $e^T \Sigma e$ per episode collected, because the difference of the pair is a pure e -displacement with every nuisance direction cancelled. Independently sampled episodes spend their variance budget on nuisance directions the pretraining has already covered. Combined with Theorem 8, the same pairing buys both identifiability of the update and $1/(1-\rho)$ in evaluation sample size — the same asset twice.

independently sampled episodes



matched counterfactual pairs

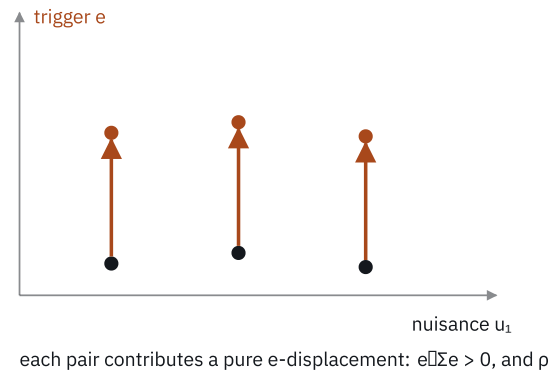


Figure 5. Theorem 10 and Corollary 10.1. Left, refined episodes drawn independently: abundant, expensive, and silent about the direction you are trying to fix. Right, the same budget spent on pairs that differ only in the trigger. The arrows are the entire information content of the dataset with respect to the repair.

THE RESULT

The eight cases, decided

Three observables per mechanism: does the probe move under the perturbation, does ablating it hurt, does injecting it transfer. Eight combinations, and the naive reading is wrong in five of them.

#	PROBE MOVES	ABLATION BITES	INJECTION TRANSFERS	LICENSED CONCLUSION	BY
1	no	no	no	Unreachable — and you could have known before running. Check the step index first (Cor. 1.1).	T1
2	yes	no	no	Innocent bystander. Downstream of the trigger, upstream of nothing that matters. Not a locus.	T2
3	no	yes	—	Flat but guilty. The failure is in the interaction with a trigger-shifted sibling. Probing can never find this; only intervention can.	T2
4	yes	yes	yes	Locus — but the <i>share</i> of the effect is ordering-dependent and meaningless until the order is declared.	T3, T3b

#	PROBE MOVES	ABLATION BITES	INJECTION TRANSFERS	LICENSED CONCLUSION	BY
5	yes	—	yes, fully	Cut set. A sufficient statistic for the trigger: $A \perp T \mid h_s$. Strongest available result. Requires episodes differing only in T .	T4
6	yes	no	yes	Redundant carrier. Null ablation is not absence. Search over sets, not units.	T5
7	no	yes	yes	Instrument blind. Two distinct defects live here: a permutation-invariant readout cannot see <i>where</i> , and a modality-mixed one attenuates toward the null. Re-probe per-modality, equivariantly and at several prefix lengths before concluding anything. A null is weak evidence at best; a positive is conservative.	T6, T6b, T6c
8	yes	no	no	Shortcut. The channel is represented and unused, because the training data made it redundant. A data problem, not a model problem — and the only case retraining fixes cheaply.	T9

Cases 3, 6 and 7 are three different reasons a mechanism can look innocent and not be. Case 2 is the reason a mechanism can look guilty and not be. Getting from “the policy failed when I moved the trigger” to “mechanism m is why” means excluding all four, and the exclusions need different experiments — which is the practical content of everything above.

APPLYING IT

The protocol on your stack

- 1 Prune statically, and declare the step index.** Write down which sources the trigger enters at, take descendants, and record whether the claim is for the first forward pass or the closed loop. For robot-initial-state this eliminates the VL pathway at $h=1$ and nothing at $h \geq 2$. T1, Cor. 1.1 — costs nothing
- 2 Read the index structure before intervening.** N1.7 pre-separates language and vision by block parity and phase: $B_L = \{0, 4, \dots, 28\}$, $B_V = \{2, 6, \dots, 30\}$, odd

blocks empty. Capture per-block DiT hidden states —

`AlternateVLDiT.forward` already accepts `return_all_hidden_states` and returns all 33 — and the pathway question is answered by observation.

Structural null: periodicity at exactly the architectural phase. **Cor. 4.1** — one capture run, no intervention

3 Fix the readout before spending episodes — two defects, one tap. Pool image and text tokens *separately* using the `image_mask` already in scope, and add the spatial moment of (6.2) beside the existing pooled roles; keep per-token activations for any role you intend to run an SAE on. Until both are done, a VL null is uninterpretable against spatial triggers and attenuated by an episode-varying mixture ratio. **T6, Cor. 6.1, T6b** — a tap change, no GPU time

4 Balance the categories, then collect matched pairs. A draw whose target category is $n=2$ cannot test a claim about that category at any sample size — stratify on the LIBERO-Plus category first, then pair within it: same seed, same initial state, one trigger toggled. Log per-episode outcomes so McNemar is available. Size from (8.1): ~ 284 paired episodes for a $2\times$ discordance ratio at 80% power. **T1** sets the strata; **T8, Cor. 10.1** size them

5 Inject only within a matched pair, on a seeded path. Theorem 4 requires source and target to differ only in the trigger, and Theorem 3 requires the run to be a function — so fix ε across both arms and every patched run, and assert bitwise reproducibility before reading any increment. Cross-*task* injection violates the premise — which the paper states as its own confound, that injected activations can produce invalid internal states through temporal misalignment. Under an L -Lipschitz downstream map, an off-manifold displacement δ bounds the measured effect by $L\delta$: an upper bound, not a path effect. **T4** + the paper's §5

6 Ablate over sets, and pre-register the order. A single-unit sweep returning zeros is Theorem 5, not a finding. Pick one topological ordering for the patching curve, declare it before running, and report Shapley only where n is small enough to afford it. **T3b, T5**

7 Check the data before the retrain. For the trigger you intend to repair, estimate Δ_x on the refined set by fitting action regressors with and without

the channel. $\Delta_x \approx 0$ means the data cannot teach the dependency, however many episodes it holds — and check that the pairs actually displace along the trigger direction. T9, T10 — do this before booking the GPU

8

Attribute at the step level, and only before divergence. Score the divergence step, not the episode outcome: episode success is a T^2 -amplified, saturating readout that will rank long-horizon suites worst for reasons unrelated to mechanism, and any probe pooled over forwards after h^* is measuring the consequence of the failure rather than its cause. T7, Cor. 7.1

Pre-registered predictions, with their falsifiers

Each follows from a theorem above plus one measured anchor. Written here so they cannot be revised after the run.

PREDICTION	FROM	STATUS
Failures carry more forwards per episode than successes, so the all-forwards statistic is unbalanced.	T6c, T7	CONFIRMED — every failure exactly 18 (the cap), successes mean 7.77, $p=1.6e-8$.
state_encoded 's 12.95× collapses toward 1× under prefix restriction, because it is post-failure displacement.	Cor. 7.1	REFUTED — 6.11× survives at the first forward; it rises to 12.95×, so roughly half is accumulation and half is a real early signal. The theorem holds; the prediction over-read it.
Per-index separation at forward 0 is not significant.	Cor. 7.1	REFUTED — 6.11×, $p=0.046$, marginal and among many tests.
The VL signals stay flat at every prefix length, for the reasons in Theorems 1, 6 and 6b.	T1, T6, T6b	REFUTED — v1_adapted_mean separates at 5.30× ($p=.0089$) at $k=2$ and decays to 1.08× by $k=18$. The dominant defect was temporal averaging, which the framework did not cover until Theorem 6c was added after the fact.
Per-block DiT readout shows language structure at blocks $\equiv 0 \bmod 4$ and visual at even $\equiv 2 \bmod 4$, odd blocks neither.	Cor. 4.1	OPEN — needs the per-block capture. Any other phase is a readout bug or a misread forward pass, not a discovery.

PREDICTION	FROM	STATUS
In a category-balanced re-draw, per-modality and equivariant VL readouts separate camera-viewpoint failures while robot-initial-state stays flat.	T1, T6, T6b	OPEN — R-036's draw has n=2 camera episodes, so this has never been tested.
v1_adapted_mean separates at k=2 on a fresh capture.	T6c	OPEN — pre-registered. The confirmation test for the exploratory result above.
Retraining on trigger-varying <i>paired</i> data improves trigger-conditional success; an equal budget of independently sampled episodes does not.	T10, Cor. 10.1	OPEN — the end-state test, and the strongest available evidence that a mechanism was repaired rather than data merely added.

What the paper hands you off the shelf

Beyond the theory it licenses, four things are directly liftable, and three caveats travel with them.

- **The per-token SAE recipe.** TopK with $k=64$, expansion $4d-8d$, trained per token and never on pooled activations. 424 SAEs across five models, 68 of them on GR00T. Feature selection by frequency-weighted contrast, $score_f = d_f \times freq_f$, with d_f Cohen's d between concept-present and concept-absent tasks and $freq_f$ the fraction of samples where the feature is in the active top- k .
- **The four injection conditions** — null, same-scene, cross-task, cross-seed — of which only same-scene and cross-seed satisfy Theorem 4's premise. Use the other two as bounds, labelled as bounds.
- **A GR00T pathway prior.** DiT ablations cost 40–80% success; Eagle moderate; VL-SA most resilient despite the worst per-token SAE quality. That is a ranked list of where to look first, and it agrees with your `state_encoded` result.
- **Action Atlas**, the released toolkit, at CWRU-AISM/action-atlas.

CAVEATS THAT TRAVEL

The paper's GR00T rows are N1.5, and N1.7's conditioning schedule is a different graph — every cut set must be re-derived, not inherited. All of it is simulation; the authors note that fine-tuning on a target robot reshapes internal representations, so a cut set is a property of a checkpoint. And their SAE features are not guaranteed disentangled: ablation validation found features encoding general motor primitives rather than clean semantic concepts, which is Theorem 5 arriving from the other direction.

STANDING ON WHAT

Assumptions

Every theorem above is conditional. This is the full list of what each one needs, and whether your stack currently supplies it — a theorem whose premise fails is not wrong, it is simply not about your situation.

RESULT	PREMISE	ON GR00T N1.7 TODAY
T1, Cor. 1.1, T2	The computation is a finite DAG with a known trigger entry point.	Holds.
T3, T3b	$F(S)$ is a <i>function</i> : same patch set in, same action out.	Fails. The denoise loop starts from an unseeded <code>torch.randn</code> (<code>groot_n1_7.py:657</code>). Needs a seeded path; R-036 <i>relies</i> on the unseeded one, so the two modes must be separate.
T4	Source and target episodes differ only in the trigger.	Fails on the current unpaired draw. Cross-task injection violates it by construction and yields an upper bound ($\leq L\delta$), not a path effect.
Cor. 4.1	The block conditioning schedule is as read from the forward pass.	Provisional. Read, not measured, and previously misread as period 2.
T5	None beyond determinism of the circuit.	Holds; it is a possibility result.

RESULT	PREMISE	ON GR00T N1.7 TODAY
T6	The perturbation acts as an exact token permutation.	Idealisation. Real perturbations are approximate permutations, so the conclusion is attenuation, not blindness.
T6b	The mixing ratio α is independent of the class label.	Plausible, untested. If instruction length correlated with difficulty it would fail, and the bias would no longer be guaranteed toward the null.
T6c	Per-forward scores are exchangeable with equal variance; signal confined to a prefix.	An approximation. The prefix location f_0 is unknown, which is why the curve rather than any single F is the estimand.
T7	Bounded per-step cost, fixed horizon, behaviour-cloned policy.	Holds.
T8	Pairs are genuinely matched; normal approximation for the power formula.	Available once seed and initial state are held across arms. Per-episode outcomes must be logged — currently they are not.
T9	Squared loss; Δ_x read on the <i>training</i> distribution.	Holds, and is checkable before training.
T10	Mechanism linear in its input.	Proved for the linear case only; the nonlinear version is weaker and must be established empirically per site.

Provenance

T1, Cor. 1.1, T2, T3, T3b, T4 and T5 are standard causal-inference and mechanistic-interpretability material, re-derived here so the page is self-contained — not new, and a referee would call them folklore. T7 and T8 are restatements of published results, attributed below. T9 and T10 are standard arguments; the application to this paper's language result is ours. **T6, T6b, T6c and Cor. 7.1 were written for this stack** and exist nowhere else. None of it is peer-reviewed in this form, and the proofs have been checked by one author only.

What the paper supplies, and what it does not

arXiv:2603.19233 supplies *vocabulary and empirical anchors* — the four injection conditions, the pooled-versus-per-token finding, the GR00T pathway ablation

ordering, the suite tables. It supplies no theorems; its only equation is $score_f = d_f \times freq_f$. Three caveats travel with everything taken from it:

- **No number in it has been reproduced here yet.** A reproduction gate is pre-registered and in flight (2026-09-22-paper-reproduction-gate.md). Until it returns, every figure quoted from the paper on this page is an anchor on trust.
- **Their GR00T is N1.5 with a 16-block DiT; ours is N1.7 with 32,** and the hook paths differ (`action_head.diffusion_model.transformer_blocks` vs `action_head.model.transformer_blocks`). Layer indices do not transfer; at best proportions do.
- **Their LIBERO GR00T checkpoints are community finetunes —** `aractingi/libero-groot-goal` loads as a *projector-only* tune over `nvidia/GR00T-N1.5-3B` . Their GR00T rows were never evidence about `nvidia/GR00T-N1.7-LIBERO` .

Sources.

- B. Grant, X. Zhao, P. Wang. *Not All Features Are Created Equal: A Mechanistic Study of Vision-Language-Action Models*. arXiv:2603.19233 — [abstract](#) · [full text](#) · [Action Atlas toolkit](#). Accepted to the Multimodal Intelligence Workshop @ ICLR.
- S. Ross & J. A. Bagnell. *Efficient Reductions for Imitation Learning*. AISTATS 2010 — [proceedings](#). Theorem 7 restates their bound.
- L. S. Shapley. *A Value for n-Person Games*. 1953. The unique attribution satisfying efficiency, symmetry, null player and linearity — the averaging in Figure 2.
- J. Pearl. *Causality*. 2nd ed., 2009. d-separation and the do-operator, behind Theorems 1 and 4.
- L. Gao et al. *Scaling and Evaluating Sparse Autoencoders*. arXiv:2406.04093 — [abstract](#). The TopK SAE formulation the paper's recipe uses.
- Checkpoints under test: [aractingi/libero-groot-goal](#) over `nvidia/GR00T-N1.5-3B`; ours is `nvidia/GR00T-N1.7-LIBERO`.

Repo anchors. RESULTS.md R-036 with its 2026-09-22 sample-composition correction and per-category tables; FINDINGS.md F9; `vla_harness/capture/taps.py`; `vla_harness/capture/analysis.py`; `experiments/ood_prefix.py`; `groot_n1_7.py`:451/457, :657; `action_head/cross_attention_dit.py`:357–380; `third_party/LIBERO-plus/.../task_classification.json`; specs 2026-09-22-prefix-restricted-reanalysis.md and 2026-09-22-paper-reproduction-gate.md.

Every theorem here is proved from the definitions in the first section, under the premises tabulated above. Every number attributed to the paper or to a run is a measurement, marked as such, and is used as an

anchor — never as a premise. Predictions that have been tested carry their outcome, including the three that were refuted.